Automotive IP & AI Security: Risk Tracking & Automated Escalation (Jira)

A self-directed portfolio project showing how a security and IP risk register with likelihood/impact scoring, standards mapping and automated escalation can be built natively in Jira. It extends the automation-driven controls pattern of the companion AI Project Controlling (EVM) project to a security and compliance domain.

Note: This is a practice build using fictional data, created to show Jira administration, automation and dashboarding skills. It is not tied to a real client or employer engagement.

Jira space: Automotive IP & AI Security (key: AIAS), team-managed Kanban. **Tools:** Jira, Jira Automation, Jira Dashboards

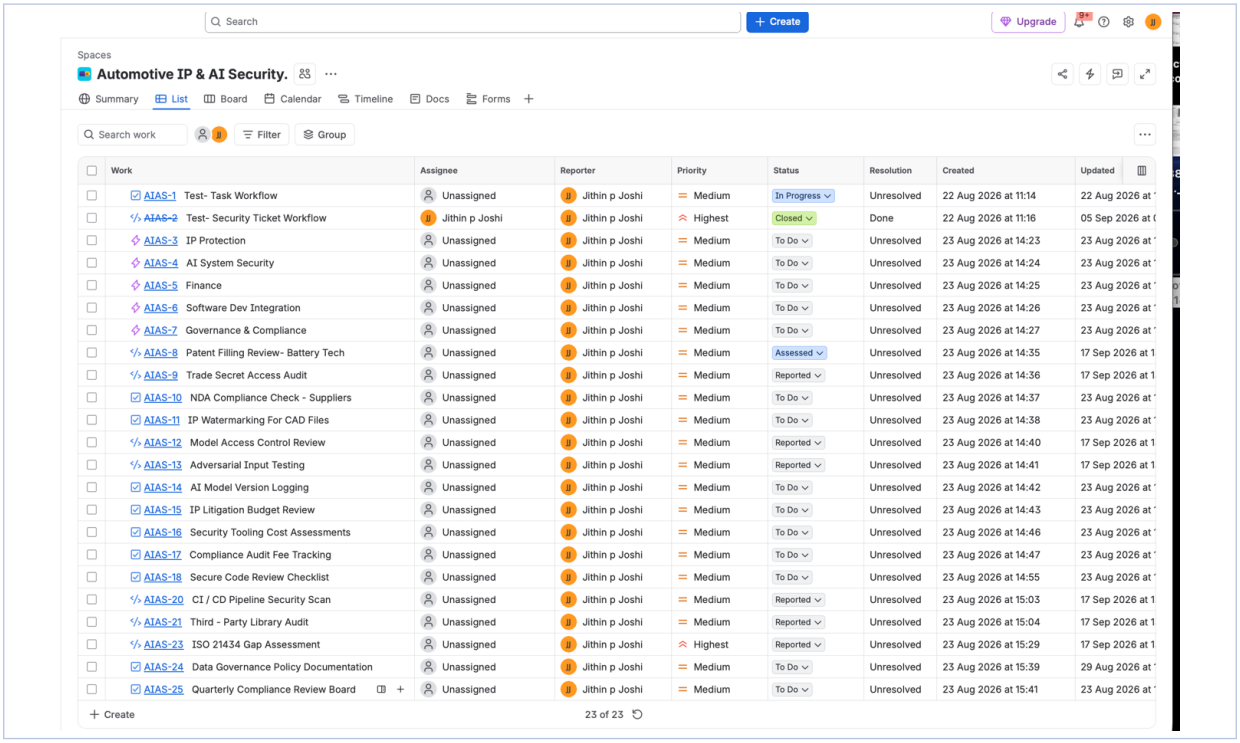
1. Problem statement

Automotive IP and AI-system security programmes must track risk across several domains (IP protection, AI model security, financial exposure, secure development and regulatory compliance) with consistent scoring and automatic escalation once a risk crosses a threshold. This project builds that register directly in Jira, using calculated risk scores and rule-based escalation instead of a separate GRC tool.

2. Epics (risk domains)

Epic	Representative tickets
IP Protection	Patent Filing Review, Trade Secret Access Audit, IP Litigation Budget Review, NDA Compliance Check, IP Watermarking
AI System Security	Model Access Control Review, Adversarial Input Testing, AI Model Version Logging
Finance, Breach Cost & Audit Trail	Compliance Audit Fee Tracking, Security Tooling Cost Assessments
Software Development Integration	CI/CD Pipeline Security Scan, Secure Code Review Checklist
Governance & Compliance	ISO 21434 Gap Assessment, Third-Party Library Audit, Data Governance Policy Documentation, Quarterly Compliance Review

23 issues across the 5 epics, mixing Tasks and a custom **Security Ticket** work type.



Full issue list: 5 epics and 23 issues across Reported, To Do, In Progress and Closed states

3. Work types and workflow

- **Task:** standard workflow To Do → In Progress → Done
- **Security Ticket** (custom work type): custom workflow below

Reported → Assessed → Mitigation In Progress → Resolved → Closed

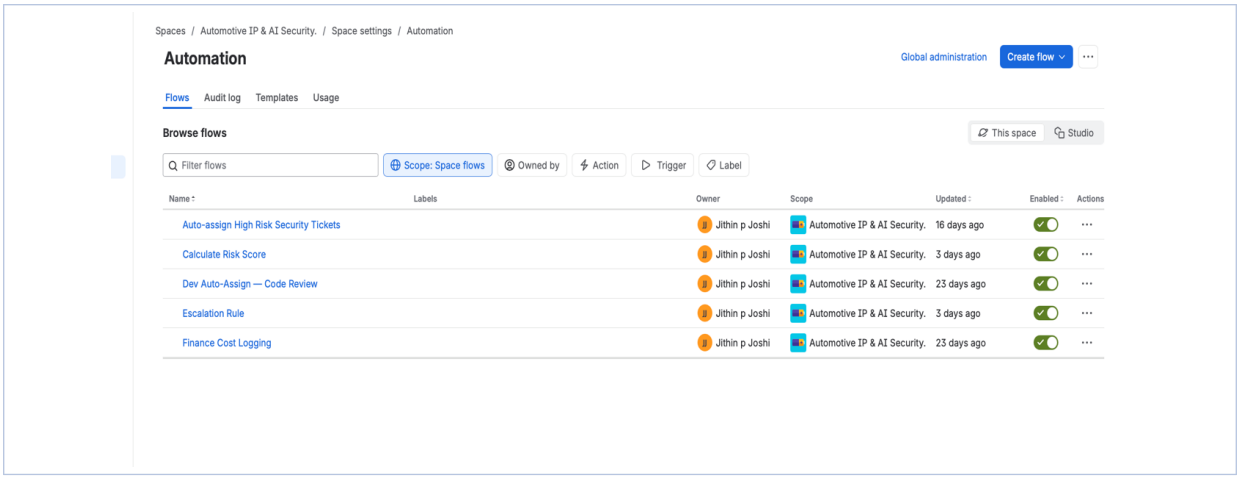
4. Custom fields

Field	Type	Purpose
Risk Category	Label	Classifies the nature of the risk
Affected System	Label	System or asset impacted
Standard Reference	Label (TISAX / ISO 21434 / GDPR)	Regulatory or standard mapping
Likelihood	Number (1–5)	Probability of occurrence
Impact	Number (1–5)	Severity if realised
Risk Score	Number (calculated)	Likelihood × Impact, set by automation
Estimated Cost	Number	Projected financial exposure
Requires Code Review (on Task)	Checkbox	Flags dev tasks that need a security review

Severity reuses Jira’s system Priority field instead of a separate custom field.

5. Automation rules

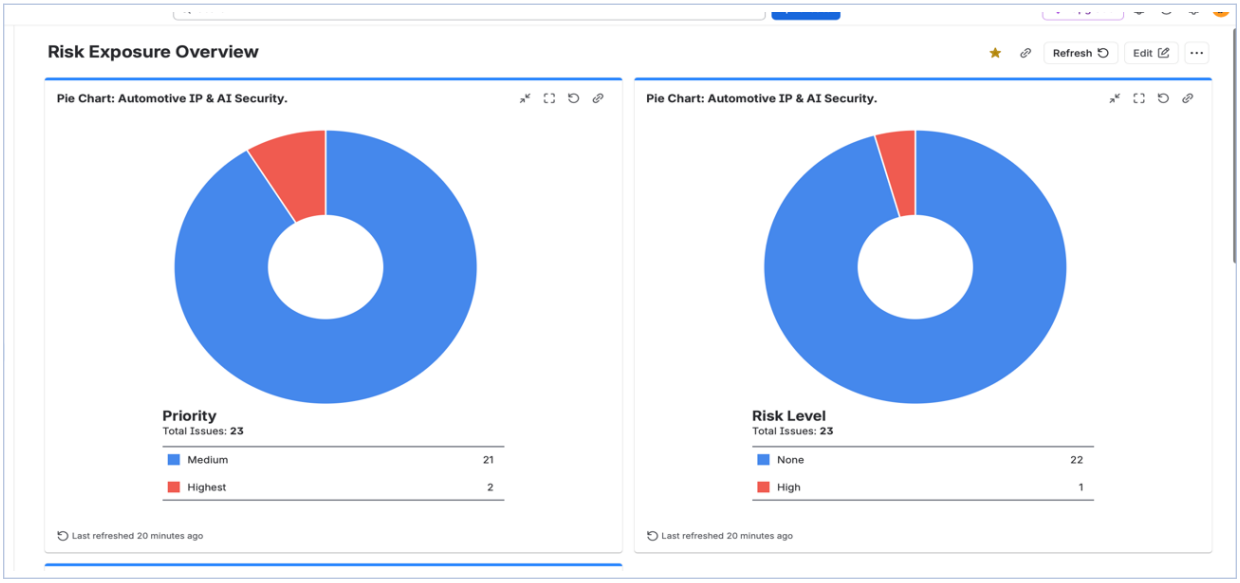
Rule	Trigger	Condition	Action
Calculate Risk Score	Likelihood or Impact changed	–	Risk Score = Likelihood × Impact
Escalation Rule	Risk Score changed	Risk Score ≥ 15	Priority → Highest, plus escalation comment
Finance Cost Logging	Status → Resolved	Affected System = Financial Analytics	Auto-comment with Estimated Cost
Dev Auto-Assign: Code Review	Task created	Requires Code Review = true	Assign, plus review checklist comment
Auto-assign High Risk Security Tickets	Risk Score changed	Risk Score ≥ threshold	Auto-assign to owner



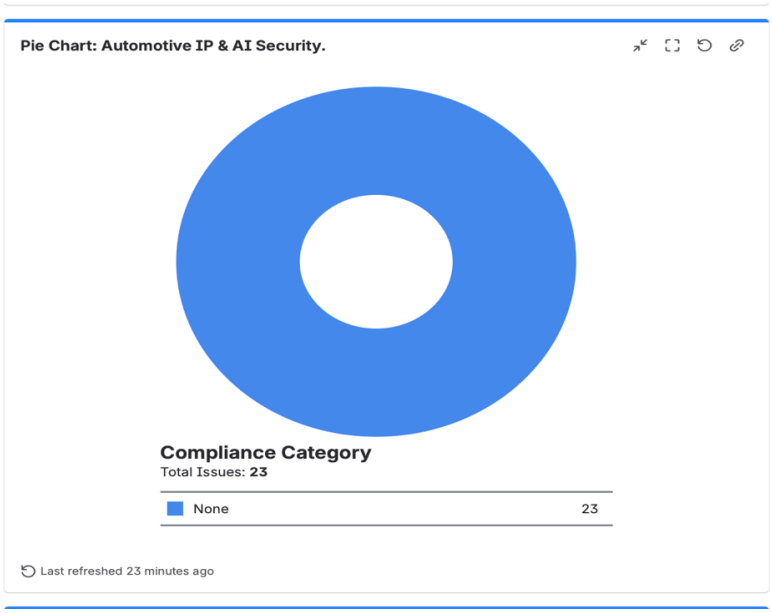
Automation rules in the space: all 5 flows owned and enabled

6. Risk Exposure Overview dashboard

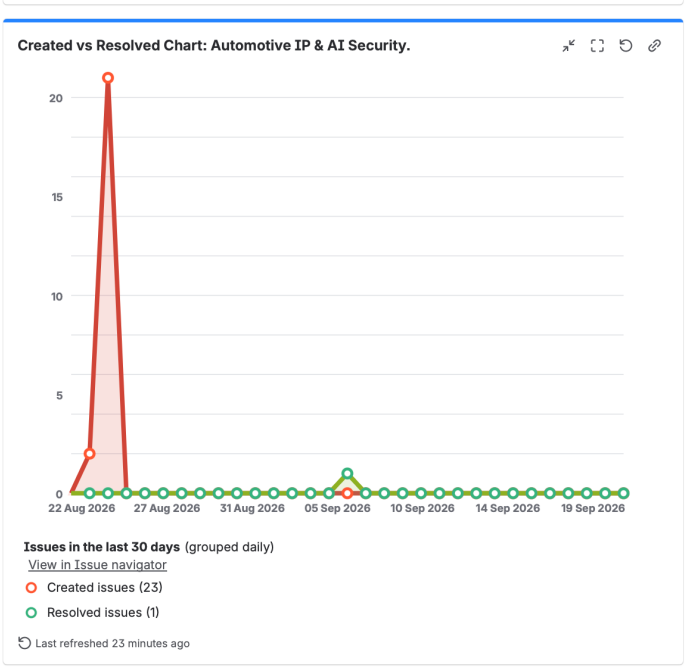
The dashboard shows the Priority breakdown, Risk Level breakdown and Compliance Category breakdown as pie charts, plus a Created vs Resolved chart.



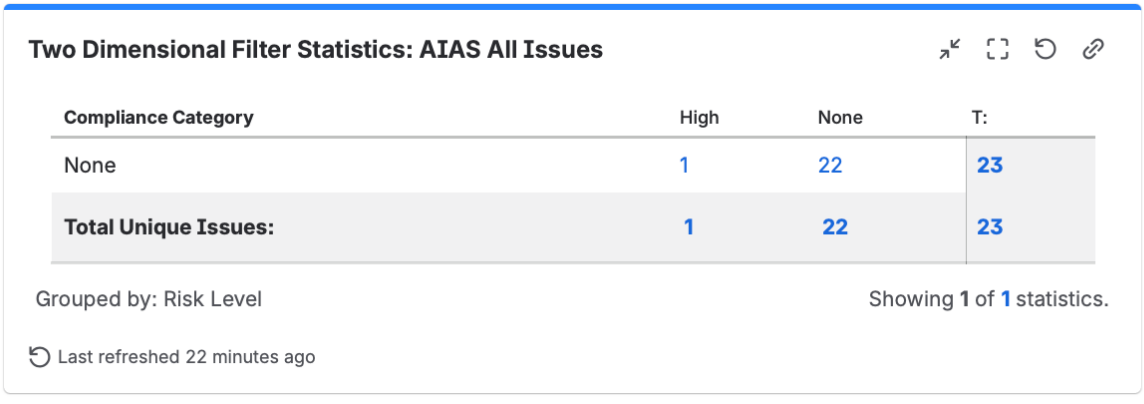
Priority and Risk Level breakdown (pie charts)



Compliance Category breakdown (pie chart)



Created vs Resolved chart



Risk Level and Compliance Category distribution once all tickets carry live Likelihood, Impact and Standard Reference values

7. Known limitation (carried forward from the EVM project)

Jira's "field value changed" trigger fires only on future edits. A value entered at ticket creation, or before the rule existed, does not retroactively trigger it. On first build most Security Tickets therefore showed no Risk Score and no Compliance Category:

- **Risk Score:** Likelihood and Impact were set once at creation, so the Calculate Risk Score rule never fired for most tickets.
- **Compliance Category (Standard Reference):** a manual label field, not automation-driven, left unset on most tickets during the initial build.

Resolution: each ticket's Likelihood and Impact were re-saved to force the Risk Score trigger, and Standard Reference was filled in manually on all Security Tickets. Catching an automation rule that silently does nothing because of trigger timing or a blank action field is documented deliberately as a real, repeatable Jira pitfall.

Tools used

- Jira (team-managed Kanban space, custom work types, custom fields, automation engine, dashboards)

What this project demonstrates

- Risk-scoring fundamentals (Likelihood × Impact matrices)
- Custom work type and workflow design in Jira
- Rule-based automation for scoring, escalation and auto-assignment
- Cross-portfolio linking between related Jira spaces
- Debugging and correcting a recurring automation trigger-timing issue

Related project

AI Project Controlling (Jira): EVM tracking and automated At Risk / Watch flagging, the companion build for this project.

Author: Jithin Pariyappanal Joshi (self-directed learning project, feedback welcome)